Programming Assignment 1

Photometric Stereo

CSE404: Introduction to Computer Vision

DGIST • Spring 2026

Outline

Motivation & Background

Dataset

Algorithm

Expected Results

Assignment Tasks

Motivation & Background

What is Photometric Stereo?

Core idea: use multiple images under different illuminations to recover 3-D surface information from a 2-D image sequence.

- Single viewpoint, fixed camera
- n light sources $\rightarrow$ n observations per pixel
- Lambertian reflectance model

Lambertian Shading

$$I = \rho\,(\mathbf{N} \cdot \mathbf{L})$$

I - observed intensity

ρ - albedo (surface colour)

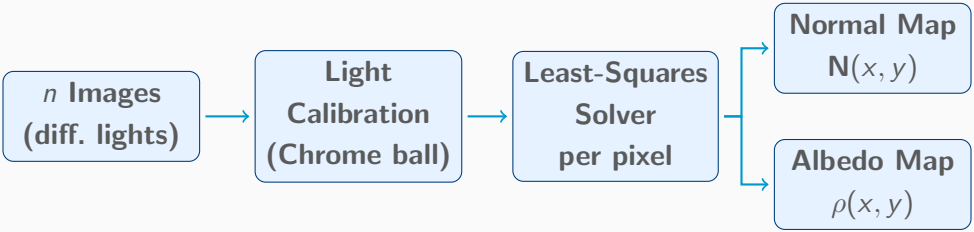
$\mathbf{N}$ - surface normal

$\mathbf{L}$ - light direction

Outputs

- **Surface normal map** $\mathbf{N}(x, y)$
- **Albedo map** $\rho(x, y)$

The Photometric Stereo System



The chrome-ball sub-system extracts light directions; the solver runs independently per pixel.

Dataset

Provided Data

Chrome Ball - `chrome.txt`

- 12 images of a mirror sphere
- `chrome.[0...11].png`
- `chrome.mask.png`
- Used to **estimate** the 12 light directions $\mathbf{L}_i$

Buddha - `buddha.txt`

- 12 images of the subject
- `buddha.[0...11].png`
- `buddha.mask.png`
- Same 12 lights as above
- Used to **recover** $\mathbf{N}$, ρ

File format (`.txt`)

`<num_images>` on line 1, then one path per line; **last line is always the mask**.

Algorithm

Step 1 — Light Direction from Chrome Ball

Sphere Surface Normal

$$\mathbf{N}(x, y) = \left(\frac{x-c_x}{r},\ \frac{y-c_y}{r},\ \sqrt{1 - n_x^2 - n_y^2}\right)^\top$$

A perfect mirror sphere obeys the **reflection equation**:

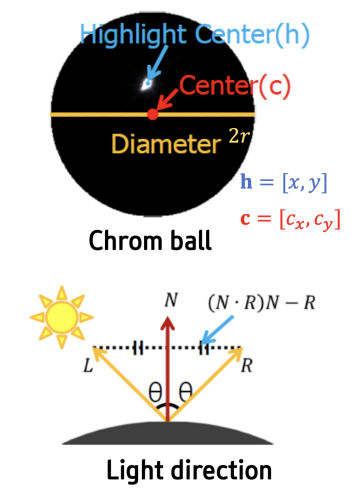
$$\mathbf{L} = 2(\mathbf{N} \cdot \mathbf{R})\mathbf{N} - \mathbf{R}$$

L: Light direction, **R**: Reflection vector

$\mathbf{R} = (0, 0, 1)^\top$ (orthographic)

h: the highlight center, c: the chrome ball center

r: the radius of the chrome ball



Step 2 — Photometric Stereo (Least Squares)

Stack all n observations per pixel:

$$\underbrace{\begin{pmatrix} I_1 \\ I_2 \\ \vdots \\ I_n \end{pmatrix}}_{\mathbf{i} \in \mathbb{R}^n} = \underbrace{\begin{pmatrix} \mathbf{L}_1^\top \\ \mathbf{L}_2^\top \\ \vdots \\ \mathbf{L}_n^\top \end{pmatrix}}_{\mathbf{L} \in \mathbb{R}^{n \times 3}} \underbrace{\rho \mathbf{N}}_{\mathbf{g} \in \mathbb{R}^3}$$

Solve with **least squares**: $\hat{\mathbf{g}} = (\mathbf{L}^\top \mathbf{L})^{-1} \mathbf{L}^\top \mathbf{i}$

Albedo

$\rho = \|\hat{\mathbf{g}}\|$

Surface Normal

$\mathbf{N} = \hat{\mathbf{g}} \,/\, \|\hat{\mathbf{g}}\|$

Expected Results

Expected Outputs

Output Files

File	Description
`normals.png`	Normal map — (N_x, N_y, N_z) encoded as RGB via $(\mathbf{N}+1)/2 \times 255$
`albedo.png`	Albedo ρ — grayscale map of intrinsic surface brightness
`shading.png`	Re-shading under frontal light $\mathbf{L} = (0,0,1)^\top$ using recovered normals

Normal Map Colour Interpretation

- Red channel $\rightarrow$ $+X$ (pointing right)
- Green channel $\rightarrow$ $+Y$ (pointing up)
- Blue channel $\rightarrow$ $+Z$ (pointing toward camera)

All normals inside the mask must satisfy $\|\mathbf{N}(x,y)\| = 1$.

Assignment Tasks

Your Tasks

What to Submit

T1 Sphere fitting — fit a circle to the chrome-ball mask to obtain (c_x, c_y, r).

T2 Normal map — compute the analytical sphere normal map.

T3 Light estimation — detect the specular highlight and apply the reflection formula to get $\mathbf{L}_i$ for all 12 lights.

T4 Photometric stereo — implement the least-squares solver and produce the surface normal and albedo maps for the Buddha.

T5 Re-shading — render the object under a novel light direction using your estimated normals.

Grading & Deliverables

Deliverables

- `pa1.py` — fully runnable Python source
- `normals.png`, `albedo.png`, `shading_frontal.png`
- `report.pdf` — brief write-up (max 2 pages)

Grading Breakdown

Task	Points
T1 Sphere fitting	10
T2 Normal map	15
T3 Light estimation	25
T4 PS solver	35
T5 Re-shading	10
Report	5
Total	**100**

Tips & Resources

Practical Tips

- Always apply the **mask** before processing
- Guard against `NaN` / divide-by-zero in unit-normalisation
- Use `np.linalg.lstsq` with `rcond=None`
- Visualise intermediate results early
- Test light directions: $\|\mathbf{L}_i\| \approx 1$ and $L_{i,z} > 0$

References

- Woodham, R.J. (1980). *Photometric method for determining surface orientation from multiple images.* Optical Engineering.
- Course lecture slides — Week 5 (Radiometry & Reflectance)
- NumPy docs: `numpy.linalg.lstsq`
- OpenCV docs: `cv2.imread`, `cv2.imwrite`

Good Luck!

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Submit via the lms website

Due: 2026.03.27 23:59PM Fri (three weeks)

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